Fairsplit Litepaper

USDC Expense Splitting with Optional Confidential Settlement

Abstract

Fairsplit is a USDC-native expense splitting app for friends, teams, and communities. It combines **onchain verifiability** with **optional confidential payments**, allowing users to coordinate shared expenses transparently or privately when needed. Fairsplit is live on **Base Sepolia** and **ARC Testnet**, with Fairblock's confidential transfer infrastructure integrated.

Why this exists

Splitting expenses sounds trivial, but current solutions force tradeoffs:

- Web2 apps are convenient, but rely on centralized custody, private databases, and trust in intermediaries.
- Onchain tools are verifiable, but often expose payment amounts and relationships publicly.
- Most tools stop at "send money" and ignore the full coordination lifecycle: reminders, status tracking, and history.
- Privacy, when available at all, is usually all-or-nothing.

Fairsplit exists because **everyday payments need coordination, verifiability, and choice not just transfers**.

Vision

Fairsplit aims to be the **default USDC coordination layer** for shared expenses:

- Simple enough for everyday use
- Verifiable through onchain state
- Privacy-optional by design
- Portable across web and mini-apps

Core Product

Fairsplit lets users:

- Create immutable USDC expense splits
- Track participant-level payment status
- Pay via **public USDC transfer** or **confidential settlement**
- Send reminders and payment requests
- Verify payments via explorer links
- Use wallet-native identity and name resolution

Key Differentiator: Privacy Is Optional

Most bill-split apps force everything to be public.
Fairsplit doesn't.

Users can choose:

- **Public flow** → standard ERC-20 USDC transfer
- **Confidential flow** → Fairblock StableTrust–powered private settlement

Same bill. Same UX. **Different privacy guarantees.**

Feature Snapshot

- **USDC-only settlement** (6-decimal precision, exact accounting)
- **Participant-level paid state tracking**
- **Confidential wallet flows** (deposit, transfer, withdraw)
- **Multi-chain support** (Base Sepolia, ARC Testnet)
- **ENS / Basename resolution**
- **Activity feed with chain badges & explorer links**
- **Mini-app compatible UX**
- **Builder attribution support**

Architecture overview

Onchain

- Smart contracts manage bill state, participant eligibility, and settlement status
- USDC-only accounting simplifies logic and expectations
- Onchain state is the single source of truth

Frontend

- Built with Next.js and React
- Wallet interactions via wagmi and viem
- Chain-aware UI with explorer links for verification

Offchain (V1)

- Profiles, friends, and activity stored in browser local storage
- Email and in-app notifications for coordination only

Why this matters

Fairsplit shows that everyday financial coordination can happen onchain **without forcing users to give up privacy or usability**. By combining USDC-native settlement, optional confidentiality, and simple coordination tools, it offers a practical foundation for social payments in Web3.

The current testnet release is a proving ground — not a finish line.

Limitations (V1)

- User data is device-specific (local storage)
- No centralized accounts or recovery
- Testnet performance may vary
- Confidential UX depends on SDK and network execution timing

These constraints are explicit and intentional for an early release.

Security & Assumptions

- Contracts are immutable post-deployment
- Confidential flows depend on correct key management
- Name resolution may fail under RPC edge cases
- Local frontend state can be cleared by users

Mitigated via explicit chain labels, explorer links, and strict validation.

Roadmap

- Mainnet rollouts
- Better error decoding
- Improved analytics
- Automated reminders

Disclaimer

Fairsplit is experimental software. Users should evaluate smart contract, wallet, and privacy tradeoffs before transacting meaningful value. This document is not financial or legal advice.